GAUSSIAN DOMINATION AND STRUCTURAL PROPERTIES OF CONTINUUM LIMITS

Abstract. Gaussian domination is preserved under weak convergence of laws of tempered fields. Under a common L^2 Gaussian bound, the generating functions of every finite family of Schwartz pairings converge locally uniformly, as do their logarithms near the origin. Consequently all joint moments and cumulants converge. We also prove a lattice compactness theorem under the weaker assumption of a uniform second-moment bound. Cell averaging and Hermite coefficient estimates give subsequential continuum laws. We prove that approximate linear identities, geometric symmetries and reflection inequalities pass to these limits, and establish tensor Schwinger-function bounds. For chamber QED3, the resulting laws are closed, translation and cubically invariant, and reflection positive at the nine lattice mirror directions at every height. We also recover the real lattice law associated with a prescribed fixed-spacing volume limit.

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1. Introduction

Let $\mathcal{S} = \mathcal{S}(\mathbb{R}^d;\mathbb{R}^r)$ be the real Schwartz space and $\mathcal{S}'$ its continuous dual. Write e_α for the standard basis of $\mathbb{R}^r$. For the normalized real Hermite functions h_k, $k \in \mathbb{N}_0^d$, set

$$(1 - \Delta + |y|^2)h_k = \lambda_k h_k, \qquad \lambda_k = 2|k|_1 + d + 1, \qquad |k|_1 = k_1 + \cdots + k_d.$$

For $t > 0$ we use the separable Hilbert space

$$H_{-t} = \{F \in \mathcal{S}' : \|F\|_{-t} < \infty\}, \qquad \|F\|_{-t}^2 = \sum_{\alpha=1}^{r} \sum_{k\in\mathbb{N}_0^d} \lambda_k^{-2t} |F(h_k e_\alpha)|^2. \tag{1}$$

All weak convergence of probability laws on this space refers to its norm topology.

Extend each real distribution complex linearly to $\mathcal{S}_\mathbb{C} = \mathcal{S}(\mathbb{R}^d;\mathbb{C}^r)$. For a law P, its smeared n-point correlation is $S_n^P(f_1,\dots,f_n) = \mathbb{E}_P \prod_{j=1}^n F(f_j)$ whenever this expectation exists.

Theorem A (Gaussian domination and convergence of correlations)**.** Fix integers $d, r \geq 1$ and a real number $t > 0$. Let P_j and P be Borel probability laws on H_{-t} such that $P_j \Longrightarrow P$. Suppose

$$\mathbb{E}_{P_j} e^{F(f)} \leq e^{\|f\|_2^2/2} \qquad (j \geq 1,\ f \in \mathcal{S}). \tag{2}$$

Then

$$\mathbb{E}_P e^{F(f)} \leq e^{\|f\|_2^2/2} \qquad (f \in \mathcal{S}). \tag{3}$$

For every finite family $f_1, \dots, f_m \in \mathcal{S}_{\mathbb{C}}$, the functions

$$Z_j(z) = \mathbb{E}_{P_j} \exp\left(\sum_{a=1}^{m} z_a F(f_a)\right), \qquad Z(z) = \mathbb{E}_P \exp\left(\sum_{a=1}^{m} z_a F(f_a)\right)$$

are entire on $\mathbb{C}^m$. They satisfy $Z_j \to Z$ locally uniformly, with every complex derivative. In particular, for every $n \geq 1$ and $f_1, \dots, f_n \in \mathcal{S}_{\mathbb{C}}$,

$$S_n^{P_j}(f_1, \dots, f_n) \longrightarrow S_n^{P}(f_1, \dots, f_n). \tag{4}$$

On the common domain

$$D_f = \left\{ z \in \mathbb{C}^m : \sum_{a=1}^{m} |z_a| \|f_a\|_2 < 2/5 \right\}, \tag{5}$$

all Z_j and Z satisfy $|Z_j - 1| < 1/2$ and $|Z - 1| < 1/2$. Their principal logarithms converge locally uniformly on D_f, with every derivative. Thus all joint cumulants converge, where

$$\kappa_n^P(f_1, \dots, f_n) = \partial_{z_1} \cdots \partial_{z_n} \log \mathbb{E}_P \exp\left(\sum_{a=1}^{n} z_a F(f_a)\right)\Bigg|_{z=0},$$

and $\kappa_n^{P_j}$ is defined in the same way. ■□

The exponential estimate controls the unbounded observables in (4). More precisely, it gives a uniform second-moment bound for each exponential in Z_j, so weak convergence passes these expectations to the limit. Local bounds for the entire generating functions then pass all their derivatives. The bound $|Z_j - 1| < 1/2$ on D_f permits the same argument for their logarithms. We give the proof in section 2, following the random-linear-form estimates and convergence argument in [Cur, Sections 5.1–5.2 and 6.2].

The weakly convergent subsequences needed for this result can be obtained from second moments alone. We state the lattice form separately, since it does not require Gaussian domination.

Theorem B (Continuum limits of lattice fields)**.** Fix integers $d, r \geq 1$, offsets $t_1, \dots, t_r \in [0,1]^d$, and a set $I \subset (0,1]$ with 0 in its closure. For each $a \in I$, let ν_a be a Borel probability law on $\mathbb{R}^{\mathbb{Z}^d \times \{1,\dots,r\}}$ with the coordinate product topology. Suppose there is a constant $0 \leq C < \infty$ such that

$$\mathbb{E}_{\nu_a} \left| \sum_{\alpha,x} b_\alpha(x) u_\alpha(x) \right|^2 \leq C \sum_{\alpha,x} |b_\alpha(x)|^2 \tag{6}$$

for every $a \in I$ and every finitely supported real coefficient array b. Define

$$(F_a)_\alpha(y) = a^{-d/2} u_\alpha(x) \quad \text{for } y \in Q_{a,\alpha,x} := a(x + t_\alpha + [-1/2, 1/2)^d). \tag{7}$$

Then F_a is almost surely a real tempered distribution, and

$$\mathbb{E}_{\nu_a} |F_a(f)|^2 \leq C \|f\|^2_{L^2(\mathbb{R}^d;\mathbb{R}^r)} \qquad (f \in \mathcal{S}). \tag{8}$$

For every fixed $t > d/2$, these fields have Borel realizations in H_{-t} whose laws are uniformly tight. Consequently, for every sequence $a_j \in I$ tending to zero there are a subsequence a_{j_ℓ} and a Borel probability measure μ on H_{-t} such that

$$\mathcal{L}(F_{a_{j_\ell}}) \implies \mu.$$

Along this subsequence the pairings with every finite family of Schwartz tests converge jointly in distribution to their pairings under μ. ■□

The hypothesis bounds the raw second moment of every finite linear combination. Bounds on the individual coordinates suffice to realize the step field, but the full inequality (6) is needed for (8). No compatibility between the laws at different spacings is assumed. The conclusion is subsequential existence; it does not assert uniqueness or that a limit is nonzero.

There are two points in the proof. First, the cell coefficients

$$b_{a,\alpha}(x) = a^{-d/2} \int_{Q_{a,\alpha,x}} f_\alpha(y)\,\mathrm{d}y \tag{9}$$

satisfy $\|b_a\|_{\ell^2} \le \|f\|_2$. Applying the lattice bound to finite truncations therefore controls the smeared field uniformly in spacing. In proposition 3.1, a spatially weighted lattice norm identifies these limits, for all tests at once, with the samplewise step distribution. Second, the test estimate gives

$$\mathbb{E}\|F_a\|_{-s}^2 \le Cr \sum_k \lambda_k^{-2s} < \infty \qquad (s > d/2).$$

For $t > s$, a closed bounded ball in H_{-s} is compact in H_{-t}: the extra factor $\lambda_k^{-2(t-s)}$ makes its coefficient tails uniformly small. Markov's inequality and Prokhorov's theorem then give the asserted subsequences, as proved in proposition 4.1. Choosing $d/2 < s < t$ gives every exponent claimed in the theorem; its proof concludes section 4.

This proof uses the Hermite compactness argument for curvature laws in [Cur, Sections 5.4 and 6.2], together with the weighted lattice realization in [Lat, Sections 5.3 and 6.1]. We isolate the second-moment assumptions used by those arguments. The dimension and the number of components enter only through the lattice weight, cell volume and Hermite multiplicities. The Hermite description of Schwartz space and its dual is recalled from [RT07, Sections 2, 4 and 5]; the probability compactness result is Prokhorov's theorem in the form stated in [Sor24, Theorem 1.1]. The application in section 7 uses the already constructed real-curvature laws of chamber QED3. The next theorem supplies the identities, symmetries and reflection forms retained by these limits.

For the structural assertion, identify the component space $\mathbb{R}^r$ with a real inner-product space V, and let W be another finite-dimensional real inner-product space. Fix an orthogonal representation $\rho : O(d) \to O(V)$. For $g(x) = Qx + b$, use the actions

$$(g \cdot f)(x) = \rho(Q) f(Q^{-1}(x - b)), \qquad (g \cdot F)(f) = F(g^{-1} \cdot f). \tag{10}$$

Invariance concerns the law as a random tempered distribution, with the sigma-algebra generated by evaluations. For an open set O, write Σ_O for the sigma-algebra generated by $F(f)$, $f \in C_c^\infty(O; V)$ real. For an open half-space H^+ and an r-invariant law P, where r reflects in its boundary, reflection positivity means

$$\mathbb{E}_P[\overline{B(r \cdot F)} B(F)] \ge 0 \qquad (B \in L^2(\Sigma_{H^+}, P)).$$

Theorem C (Identities, symmetries and reflection positivity)**.** Let $P_j \Rightarrow P$ weakly on H_{-t}, and suppose

$$\mathbb{E}_{P_j} e^{F(f)} \le e^{\|f\|_2^2/2} \qquad (j \ge 1,\ f \in \mathcal{S} \text{ real}). \tag{11}$$

Let $D^* : C_c^\infty(\mathbb{R}^d; W) \to \mathcal{S}$ be continuous and linear. Let $\mathcal{G}$ be a group of Euclidean isometries containing all translations and a finite set $\mathcal{R}$ of reflections through the origin. Choose an open side H_r^+ for each $r \in \mathcal{R}$. Assume:

(i) For each $\xi \in C_c^\infty(\mathbb{R}^d; W)$ there are real $v_j(\xi) \in \mathcal{S}$ such that $F(v_j(\xi)) = 0$, P_j-almost surely, and $\|v_j(\xi) - D^*\xi\|_2 \to 0$.

(ii) For each $g \in \mathcal{G}$ there are real linear maps $T_{j,g} : \mathcal{S} \to \mathcal{S}$ satisfying

$$\mathbb{E}_{P_j} e^{iF(T_{j,g} f)} = \mathbb{E}_{P_j} e^{iF(f)}, \qquad \|T_{j,g} f - g \cdot f\|_2 \longrightarrow 0 \quad (f \in \mathcal{S}).$$

(iii) For every $r \in \mathcal{R}$, every finite family of real $f_a \in C_c^\infty(H_r^+; V)$, and every $z_a \in \mathbb{C}$,

$$\sum_{a,b} \overline{z}_a z_b \mathbb{E}_{P_j} e^{iF(f_b - r\cdot f_a)} \geq 0 \tag{12}$$

for all sufficiently large j, depending on the family.

Then P satisfies $\mathbb{E}_P e^{F(f)} \leq e^{\|f\|_2^2/2}$ for every real $f \in \mathcal{S}$, is $\mathcal{G}$-invariant, and almost every F satisfies $F(D^*\xi) = 0$ simultaneously for all ξ. The law is reflection invariant and reflection positive at every translate of every plane in $\mathcal{R}$. Along the same sequence, all joint moments and cumulants of every fixed finite family of Schwartz evaluations converge to those of P. ■□

A linear identity is retained because its error vanishes in mean square. Symmetry is expressed by characteristic functions, whose change is bounded by the L^2 error in the transformed test. Reflection positivity passes through a finite sum of bounded continuous characters; reflection invariance then extends the form to the positive-side L^2 space. These are the structural limit arguments of [Cur, Sections 6.3 and 6.5], stated here with their test-approximation hypotheses explicit. Their proofs are given in section 5.

Gaussian stability and convergence of moments and cumulants are provided by theorem A. The structural arguments require only $\mathbb{E}|F(f)|^2 \leq C\|f\|_2^2$, except that passage of the reflection form requires no moment estimate. The common exponential hypothesis supplies the quadratic control and the correlation conclusions.

The application in section 7 starts from specified infinite-lattice laws. Their existence, reflection inequalities and estimates are supplied by [Ch, Theorem A]. The fixed-spacing identification with prescribed volume limits is proved in section 7.4, following [Cur, Section 6.4]. The resulting symmetry group is the affine cubic group; the limit argument introduces no additional rotations.

2. Gaussian bounds and convergence of correlations

We first collect the estimates for one law. Their proofs are the random-linear-form arguments of [Cur, Proposition 5.1 and Section 5.2]. They apply in every finite dimension and for any finite number of components, since they use only the L^2 norm of the tests.

Proposition 2.1. Let Q be a probability law of real tempered distributions with measurable test pairings, and suppose

$$\mathbb{E}_Q e^{F(f)} \leq e^{\|f\|_2^2/2} \qquad (f \in \mathcal{S}). \tag{13}$$

Then, for real f,

$$\mathbb{E}_Q F(f) = 0, \qquad \mathbb{E}_Q |F(f)|^2 \leq \|f\|_2^2. \tag{14}$$

For $p > 0$ set $c_p = 2\,2^{p/2}\Gamma(1 + p/2)$. One has

$$\mathbb{E}_Q |F(f)|^p \leq c_p \|f\|_2^p \qquad (f \in \mathcal{S}).$$

For complex tests $f_1, \dots, f_n$,

$$\mathbb{E}_Q \prod_{a=1}^n |F(f_a)| \leq 2^{n/2} c_n \prod_{a=1}^n \|f_a\|_2. \tag{15}$$

The generating function of each finite family of complex tests is entire. For every $g \in \mathcal{S}_\mathbb{C}$,

$$\left|\mathbb{E}_Q e^{F(g)}\right| \leq e^{\|\operatorname{Re} g\|_2^2/2}, \tag{16}$$

$$\left|\mathbb{E}_Q e^{F(g)} - 1\right| \leq e^{2\|g\|_2^2} - 1. \tag{17}$$

■

Proof. For real f, put $Y = F(f)$ and $v = \|f\|_2$. Applying (13) to tf gives

$$\mathbb{E}_Q e^{tY} \le e^{t^2v^2/2} \qquad (t \in \mathbb{R}),$$
$$\mathbb{E}_Q e^{t|Y|} \le 2e^{t^2v^2/2} \qquad (t \ge 0).$$

The second bound justifies differentiation at zero. Comparing the linear terms in the first bound for both signs of t gives $\mathbb{E}_Q Y = 0$; comparing the quadratic terms gives $\mathbb{E}_Q Y^2 \le v^2$. If $v = 0$, this also gives $Y = 0$ almost surely. If $v > 0$, Markov's inequality with $t = u/v^2$, applied to Y and $-Y$, gives

$$Q(|Y| \ge u) \le 2e^{-u^2/(2v^2)} \qquad (u > 0).$$

Integration yields

$$\mathbb{E}_Q|Y|^p = p\int_0^\infty u^{p-1}Q(|Y| > u)\,\mathrm{d}u \le 2\,2^{p/2}\Gamma(1 + p/2)v^p.$$

For $g = g_1 + ig_2$ with real g_1, g_2 and integer $n \ge 1$, the triangle inequality in $L^n(Q)$ gives

$$\|F(g)\|_{L^n(Q)} \le c_n^{1/n}(\|g_1\|_2 + \|g_2\|_2) \le \sqrt{2}\,c_n^{1/n}\|g\|_2.$$

Hölder's inequality proves (15).

On a compact polydisc in $\mathbb{C}^m$, the integrand $\exp(\sum_a z_a F(f_a))$ is bounded in absolute value by

$$\exp\left(C\sum_a(|F(\operatorname{Re} f_a)| + |F(\operatorname{Im} f_a)|)\right)$$

for a constant C depending only on that polydisc. Hölder's inequality and the exponential bound above make this majorant integrable. The same bound on a larger polydisc justifies differentiation under the expectation, so the generating function is entire. Taking the modulus of its integrand and applying (13) proves (16).

For the last estimate, put $\varphi(u) = e^u - 1 - u$. Since $\mathbb{E}_Q F(g) = 0$ and $|e^z - 1 - z| \le \varphi(|z|)$,

$$\left|\mathbb{E}_Q e^{F(g)} - 1\right| \le \mathbb{E}_Q\varphi(|F(g)|).$$

For $u \ge 0$, $\varphi(u) \le 2(\cosh u - 1)$. Writing $g = g_1 + ig_2$, convexity and monotonicity on $[0, \infty)$ therefore give

$$\begin{aligned}\mathbb{E}_Q\varphi(|F(g)|) &\le \tfrac{1}{2}\mathbb{E}_Q\varphi(2|F(g_1)|) + \tfrac{1}{2}\mathbb{E}_Q\varphi(2|F(g_2)|)\\ &\le e^{2\|g_1\|_2^2} - 1 + e^{2\|g_2\|_2^2} - 1\\ &\le e^{2\|g\|_2^2} - 1.\end{aligned}$$

This proves (17). □

Proof of Theorem A. Schwartz evaluation is continuous on H_{-t}: $|F(f)| \le \|F\|_{-t}\|f\|_t$, with the test norm defined in the proof of proposition 4.1. For real f, the function $F \mapsto \min\{R, e^{F(f)}\}$ is bounded and continuous. Weak convergence gives

$$\mathbb{E}_P \min\{R, e^{F(f)}\} = \lim_{j\to\infty} \mathbb{E}_{P_j} \min\{R, e^{F(f)}\} \le e^{\|f\|_2^2/2}.$$

Letting $R \to \infty$ proves (3). Thus proposition 2.1 applies to P as well as to every P_j. In particular, the generating functions in the theorem are entire.

Fix $f_1, \dots, f_m \in \mathcal{S}_\mathbb{C}$ and write $g_z = \sum_a z_a f_a$. The map $F \mapsto \exp(F(g_z))$ is continuous, and

$$\mathbb{E}_{P_j}\left|e^{F(g_z)}\right|^2 = \mathbb{E}_{P_j} e^{2F(\operatorname{Re} g_z)} \le e^{2\|\operatorname{Re} g_z\|_2^2}. \tag{18}$$

The same inequality holds under P. For each z, it gives uniform integrability of these exponentials. Truncating their real and imaginary parts, passing the bounded continuous expectations to the weak limit, and removing the truncations therefore proves $Z_j(z) \to Z(z)$.

The right side of (16), applied to g_z, is bounded on each compact subset of $\mathbb{C}^m$, uniformly in j. Cauchy's estimates on a larger polydisc give uniform bounds for the first derivatives on a smaller one. The functions Z_j are consequently equicontinuous on compact sets. A finite covering by small balls, together with pointwise convergence and continuity of Z, upgrades the convergence to local uniform convergence. Cauchy's formula on surrounding polydiscs then gives local uniform convergence of every complex derivative. Differentiation under the expectations identifies the derivatives at zero with joint moments, proving (4).

For $z \in D_f$, the triangle inequality gives $\|g_z\|_2 \le \sum_a |z_a| \|f_a\|_2 < 2/5$. Hence (17) gives

$$|Z_j(z) - 1|,\ |Z(z) - 1| \le e^{2\|g_z\|_2^2} - 1 < e^{8/25} - 1 < \tfrac{1}{2}.$$

All the images lie in the disk of radius $1/2$ about 1, on which the principal logarithm is analytic. Local uniform convergence of Z_j gives local uniform convergence of their logarithms on D_f. Cauchy's formula again gives convergence of every derivative, and in particular of every joint cumulant. □

3. Realization of the step field

We first work at one fixed spacing. The estimate that makes the step field a distribution may depend on this spacing. The estimate on its second moments will retain the constant in (6).

Proposition 3.1. Fix $d, r \ge 1$, $a > 0$ and offsets $t_\alpha \in [0,1]^d$. Suppose a Borel probability law ν on real lattice arrays satisfies (6) with constant C. On one measurable set of full measure, the step function (7) is tempered and, for every $f \in \mathcal{S}$,

$$F_a(f) = \sum_{\alpha,x} b_{a,\alpha}(x) u_\alpha(x), \qquad \sum_{\alpha,x} |b_{a,\alpha}(x) u_\alpha(x)| < \infty. \tag{19}$$

After setting $F_a = 0$ off this set, every test pairing is measurable. The finite truncations of (19) converge in $L^2(\nu)$, and

$$\mathbb{E}_\nu |F_a(f)|^2 \le C \|b_a\|_{\ell^2}^2 \le C \|f\|_2^2.$$

■

Proof. Choose $p > d/2$ and put

$$\|u\|_{\ell^2_{-p}}^2 = \sum_{\alpha,x} (1 + |x|^2)^{-p} |u_\alpha(x)|^2.$$

Apply (6) to each coordinate. Tonelli's theorem gives

$$\mathbb{E}_\nu \|u\|_{\ell^2_{-p}}^2 \le Cr \sum_{x \in \mathbb{Z}^d} (1 + |x|^2)^{-p} < \infty. \tag{20}$$

Thus $\Omega_p := \{u : \|u\|_{\ell^2_{-p}} < \infty\}$ is a measurable set of full measure, independent of the test function.

For $u \in \Omega_p$, weighted Cauchy–Schwarz yields

$$\sum_{\alpha,x} |b_{a,\alpha}(x) u_\alpha(x)| \le \|u\|_{\ell^2_{-p}} \left(\sum_{\alpha,x} (1 + |x|^2)^p |b_{a,\alpha}(x)|^2 \right)^{1/2}. \tag{21}$$

The second factor is controlled by a Schwartz seminorm. Indeed, Cauchy–Schwarz on each cube bounds its square by

$$\sum_{\alpha,x} (1 + |x|^2)^p \int_{Q_{a,\alpha,x}} |f_\alpha(y)|^2 \, \mathrm{d}y \le C_{a,p,d} \int_{\mathbb{R}^d} (1 + |y|^2)^p |f(y)|^2 \, \mathrm{d}y.$$

Here $|x| \le a^{-1}|y| + \frac{3}{2}\sqrt{d}$ on the cube, and the cubes partition $\mathbb{R}^d$ for each component. The constant may depend on a, p, d, but is uniform in the allowed offsets. The right side is a continuous Schwartz seminorm squared. Consequently (19) defines a continuous linear functional for every $u \in \Omega_p$. The same estimate with $|f_\alpha|$ in the cell integrals shows absolute integrability against the step function, so

the functional is precisely its distributional pairing. Finite coordinate sums are measurable. Their pointwise limits on Ω_p, extended by zero outside it, prove measurability of the pairings.

For the uniform estimate, use Cauchy–Schwarz without the spatial weight. Since each cube has volume a^d,

$$\sum_{\alpha,x}\left|a^{-d/2}\int_{Q_{a,\alpha,x}} f_\alpha(y)\,\mathrm{d}y\right|^2 \le \sum_{\alpha,x}\int_{Q_{a,\alpha,x}} |f_\alpha(y)|^2\,\mathrm{d}y = \|f\|_2^2. \tag{22}$$

Let $b_a^{(m)}$ be the restriction of b_a to $|x| \le m$, with every component retained. These restrictions converge to b_a in ℓ^2. Applying (6) to $b_a^{(m)} - b_a^{(n)}$ shows that their pairings form an $L^2(\nu)$ Cauchy sequence. Their almost sure limit on Ω_p is (19), so their L^2 limit agrees with $F_a(f)$. Passing to the limit in the finite quadratic bound and then using (22) proves the assertion. □

4. Compactness from Hermite coefficients

The preceding proposition reduces the problem to a family of random distributions with a common L^2 test estimate. We state the resulting compactness criterion independently of the lattice. The proof retains the coefficient argument of [Cur, Section 5.4] and [Lat, Section 6.1].

Proposition 4.1. Let $(X_i)_{i\in J}$ be a family of random real tempered distributions in $\mathcal{S}'(\mathbb{R}^d;\mathbb{R}^r)$, defined on possibly different probability spaces. Suppose $X_i(f)$ is measurable for each $f \in \mathcal{S}$ and

$$\mathbb{E}|X_i(f)|^2 \le C\|f\|_2^2 \qquad (i \in J,\ f \in \mathcal{S}) \tag{23}$$

for one $0 \le C < \infty$. For every $s > d/2$, the fields have Borel realizations in H_{-s} and

$$\sup_i \mathbb{E}\|X_i\|_{-s}^2 \le B_s := Cr\sum_{k\in\mathbb{N}_0^d} \lambda_k^{-2s} < \infty. \tag{24}$$

For every $t > s$ and $R > 0$, the set

$$K_R = \{X \in H_{-s} : \|X\|_{-s} \le R\}$$

is compact in H_{-t} and $\inf_i \mathbb{P}(X_i \in K_R) \ge 1 - B_s/R^2$. The laws are uniformly tight in H_{-t} and every sequence has a weakly convergent subsequence there. The inclusion $H_{-t} \hookrightarrow \mathcal{S}'$ is continuous for the strong distribution topology. ■

Proof. We first recall the identification of the Hilbert space. If $f_{\alpha k} = \langle f_\alpha, h_k\rangle_{L^2}$, the Hermite expansion of $f \in \mathcal{S}$ converges in $\mathcal{S}$. The norm $\|f\|_s$, defined by

$$\|f\|_s^2 := \sum_{\alpha,k} \lambda_k^{2s}|f_{\alpha k}|^2$$

is a continuous Schwartz seminorm. Indeed, for real $s \ge 0$ and an integer $m \ge s$,

$$\|f\|_s \le \|(1 - \Delta + |y|^2)^m f\|_2.$$

These facts are the Hermite description of Schwartz space in [RT07, Sections 2, 4 and 5]. For any coefficients $c_{\alpha k}$ with $\sum \lambda_k^{-2s}|c_{\alpha k}|^2 < \infty$, the formula

$$f \longmapsto \sum_{\alpha,k} c_{\alpha k} f_{\alpha k}$$

defines a tempered distribution by weighted Cauchy–Schwarz. It has the prescribed Hermite coefficients. Conversely, the Schwartz expansion identifies every $X \in H_{-s}$ with this formula for $c_{\alpha k} = X(h_k e_\alpha)$. Thus H_{-s} is isometric to a weighted ℓ^2 space and is separable and complete.

Expected norm and measurable realization. The bound (23) gives $\mathbb{E}|X_i(h_k e_\alpha)|^2 \le C$. Tonelli's theorem therefore gives (24). To check summability, group the indices by $|k|_1 = n$:

$$\sum_{k\in\mathbb{N}_0^d} \lambda_k^{-2s} = \sum_{n=0}^{\infty} \binom{n+d-1}{d-1} (2n+d+1)^{-2s} < \infty \qquad (2s > d). \tag{25}$$

The multiplicity is of order $(1+n)^{d-1}$. On the measurable full-measure event where the norm is finite, the finite Hermite sums of X_i converge in H_{-s}. Each sum is a Borel random variable, since it has finitely many measurable coefficients. Their limit, extended by zero off that event, is a Borel H_{-s}-valued random variable. The coefficient identification above shows that it agrees there with the given distribution X_i, for every Schwartz test at once.

Compact balls. For $X \in K_R$, the tail of the weaker norm satisfies

$$\sum_{\alpha,k:\lambda_k>M} \lambda_k^{-2t} |X(h_k e_\alpha)|^2 \le M^{-2(t-s)} R^2. \tag{26}$$

Take a sequence (X_n) in K_R. Each coefficient is bounded by $R\lambda_k^s$, so a diagonal subsequence makes every coefficient converge to a number $c_{\alpha k}$. Fatou's lemma gives $\sum \lambda_k^{-2s}|c_{\alpha k}|^2 \le R^2$. These coefficients therefore define an $X \in K_R$. On the finitely many indices with $\lambda_k \le M$, coefficient convergence gives convergence in H_{-t}. On the remaining indices, the squared norm of $X_n - X$ is at most $4R^2M^{-2(t-s)}$, by (26) applied to the difference. First let $n \to \infty$ and then $M \to \infty$. The chosen subsequence converges to X in H_{-t}. This proves compactness of K_R in that metric space.

Markov's inequality and (24) give $\mathbb{P}(X_i \notin K_R) \le B_s/R^2$, uniformly in i. Thus the laws are tight in the separable complete Hilbert space H_{-t}. Prokhorov's theorem [Sor24, Theorem 1.1] gives their relative weak sequential compactness.

Finally, for a bounded subset $D \subset \mathcal{S}$, weighted Cauchy–Schwarz gives

$$\sup_{f\in D} |X(f)| \le \|X\|_{-t} \sup_{f\in D} \|f\|_t. \tag{27}$$

The last supremum is finite. These are the seminorms defining the strong topology of $\mathcal{S}'$, so the inclusion is continuous. □

Proof of Theorem B. Apply proposition 3.1 to each ν_a. It gives one tempered step distribution at that spacing, measurable test pairings, and (8) with the common constant C. Fix $t > d/2$ and choose $d/2 < s < t$. Proposition 4.1 gives Borel laws on H_{-t}, uniform tightness and a weakly convergent subsequence along any prescribed sequence $a_j \to 0$. For any finite collection $f_1, \dots, f_m \in \mathcal{S}$, the map $X \mapsto (X(f_1), \dots, X(f_m))$ is continuous on H_{-t} by (27). Composing bounded continuous functions on $\mathbb{R}^m$ with this map proves the asserted joint convergence. The continuous inclusion also regards the limit as a Borel law on the space of tempered distributions. □

5. The structural limit passages

5.1. **Linear identities and approximate symmetries.** Suppose $P_j \Rightarrow P$ on H_{-t} and

$$\mathbb{E}_{P_j}|F(f)|^2 \le C\|f\|_2^2 \qquad (f \in \mathcal{S} \text{ real}) \tag{28}$$

with one finite C. Under item (i), exact annihilation of $v_j(\xi)$ gives

$$\mathbb{E}_{P_j}|F(D^*\xi)|^2 \le C\|D^*\xi - v_j(\xi)\|_2^2 \longrightarrow 0. \tag{29}$$

For $M > 0$, the function $\min\{|F(D^*\xi)|^2, M\}$ is bounded and continuous on H_{-t}. Its limiting expectation is zero, so $F(D^*\xi) = 0$, P-almost surely. Choose a countable dense family of smooth tests on each member of a compact exhaustion of $\mathbb{R}^d$, and intersect the corresponding full-measure events. Continuity of D^* and of each sample distribution extends the identity to every compact smooth test.

For symmetries, put $\chi_j(f) = \mathbb{E}_{P_j} e^{iF(f)}$. The inequality $|e^{iu} - e^{iv}| \le |u - v|$ and (28) imply

$$|\chi_j(f) - \chi_j(h)| \le \sqrt{C}\, \|f - h\|_2 \qquad (f, h \in \mathcal{S} \text{ real}). \tag{30}$$

For a fixed g in item (ii), this gives

$$|\chi_j(g \cdot f) - \chi_j(f)| \le \sqrt{C}\, \|g \cdot f - T_{j,g} f\|_2 \longrightarrow 0.$$

Weak convergence passes both fixed-test characteristic functions to the limit. Applying the equality to all real linear combinations of any finite family proves equality of its joint laws. These evaluations generate the sigma-algebra, so P is g-invariant.

In particular, suppose P_j is invariant under translations in $a_j\mathbb{Z}^d$, $a_j \to 0$. Given b, choose $b_j \in a_j\mathbb{Z}^d$ with $b_j \to b$. The maps $T_{j,\tau_b} f = \tau_{b_j} \cdot f$ meet the approximation hypothesis by continuity of translations on L^2. Exact fixed symmetries use $T_{j,g} f = g \cdot f$.

5.2. **Reflection forms.** For any law of tempered distributions, the linear span of $e^{iF(f)}$, $f \in C_c^\infty(O; V)$ real, is dense in $L^2(\Sigma_O)$. To see this, let B be orthogonal to these characters and condition it on finitely many evaluations. Orthogonality to every real linear combination of their tests says that the Fourier transform of the resulting finite complex measure is zero. Fourier uniqueness makes that conditional expectation zero. A countable dense family of compact tests generates Σ_O, since every sample is a continuous distribution. Conditional expectations on its increasing finite evaluation sigma-algebras converge in L^2, proving the claim.

Fix $r \in \mathcal{R}$. For $B(F) = \sum_b z_b e^{iF(f_b)}$, the reflected form equals the left side of (12). Each summand is bounded and continuous on H_{-t}, so weak convergence gives

$$\mathbb{E}_P[\overline{B(r \cdot F)} B(F)] \ge 0.$$

The preceding symmetry argument gives r-invariance of P. Consequently

$$|\mathbb{E}_P[\overline{B(r \cdot F)} C(F)]| \le \|B\|_{L^2(P)} \|C\|_{L^2(P)}. \tag{31}$$

Density extends positivity to all $B \in L^2(\Sigma_{H_r^+}, P)$. If the plane of r is $n \cdot x = 0$, its translate at height c has reflection

$$r_c = \tau_v r \tau_{-v}, \qquad v = c|n|^{-2} n.$$

Translation invariance identifies the two forms and carries their positive-side sigma-algebras to one another. It also gives r_c-invariance.

Proof of theorem C. The exponential bound implies $\mathbb{E}_{P_j} F(f) = 0$ and $\mathbb{E}_{P_j} |F(f)|^2 \le \|f\|_2^2$; see proposition 2.1. The preceding arguments with $C = 1$ prove the identities, invariance and reflection positivity. The Gaussian bound for P and convergence of all finite-family moments and cumulants follow from theorem A, whose hypotheses are the weak convergence and (11). □

6. Tensor Schwinger functions

The following estimate is the argument of [Cur, Section 5.3]. It applies to any Gaussian-dominated real two-form law in three Euclidean dimensions. For $n \ge 1$, set

$$\mathcal{S}_n = \mathcal{S}((\mathbb{R}^3)^n; (\Lambda^2 \mathbb{C}^3)^{\otimes n}), \qquad q_m(\Psi) = \max_{\substack{|\alpha| \le m \\ \text{components}}} \sup_y (1 + |y|^2)^{m/2} |\partial^\alpha \Psi(y)|.$$

Proposition 6.1. Let P be a law of real tempered two-forms with $\mathbb{E}_P e^{F(f)} \le e^{\|f\|_2^2/2}$ for all real Schwartz f. For each $n \ge 1$ there is a continuous linear map $\Psi \mapsto Q_\Psi$ from $\mathcal{S}_n$ to $L^2(P)$ which agrees with $\prod_a F(f_a)$ on decomposable tests. The functional $S_n(\Psi) = \mathbb{E}_P Q_\Psi$ is the unique continuous extension of the smeared moments to $\mathcal{S}_n$, and

$$|S_n(\Psi)| \le 2\, 66^n \sqrt{n!}\, q_{6n}(\Psi). \tag{32}$$

■

Proof. Let $h_\kappa = h_k e_{\ell m}$ be the real Hermite two-form basis. The eigenvalues of $A = 1 + |x|^2 - \Delta$ are $\lambda_\kappa = 2|k|_1 + 4$. Expand in Schwartz space

$$\Psi = \sum_{\boldsymbol{\kappa}} c_{\boldsymbol{\kappa}}(\Psi) h_{\kappa_1} \otimes \cdots \otimes h_{\kappa_n}.$$

Counting eigenvalues gives

$$Z_4 := \sum_\kappa \lambda_\kappa^{-4} = \frac{3}{32} \sum_{m \geq 0} \frac{m+1}{(m+2)^3} \leq \frac{3}{32} \left(\frac{\pi^2}{6} - 1 \right) < \frac{1}{16}.$$

Weighted Cauchy–Schwarz and Parseval imply

$$\sum_{\boldsymbol{\kappa}} |c_{\boldsymbol{\kappa}}(\Psi)| \leq Z_4^{n/2} \|(A^2)^{\otimes n} \Psi\|_2.$$

For $M(x) = 1 + |x|^2$,

$$A^2 = M^2 - 2M\Delta - 4x \cdot \nabla - 6 + \Delta^2.$$

The sum of the absolute differential coefficients is bounded by $34M^2$. Applying this in each argument gives

$$|(A^2)^{\otimes n} \Psi(x)| \leq 34^n q_{6n}(\Psi) \prod_{a=1}^n (1 + |x_a|^2)^{-1}$$

componentwise. Since $\int_{\mathbb{R}^3} (1 + |x|^2)^{-2} \, \mathrm{d}x = \pi^2$ and there are 3^n components,

$$\sum_{\boldsymbol{\kappa}} |c_{\boldsymbol{\kappa}}(\Psi)| \leq 4^{-n} (34\pi\sqrt{3})^n q_{6n}(\Psi). \tag{33}$$

For real tests the Gaussian moment estimate is

$$\mathbb{E}_P \prod_{a=1}^m |F(f_a)| \leq c_m \prod_a \|f_a\|_2, \qquad c_m = 2\, 2^{m/2} \Gamma(1 + m/2) \leq 2\, 2^{m/2} \sqrt{m!};$$

see proposition 2.1 and its proof. The last inequality is the midpoint log-convexity bound for the gamma function between 1 and $m + 1$. The $2n$-moment estimate on real basis elements shows that

$$Q_\Psi = \sum_{\boldsymbol{\kappa}} c_{\boldsymbol{\kappa}}(\Psi) \prod_{a=1}^n F(h_{\kappa_a}) \tag{34}$$

converges absolutely in $L^2(P)$, with norm at most $\sqrt{c_{2n}} \sum |c_{\boldsymbol{\kappa}}(\Psi)|$. This proves continuity into $L^2(P)$. Expanding each f_a in the Hermite basis and using the same product bound identifies the sum with $\prod_a F(f_a)$ on decomposable tests. Density of finite Hermite tensors gives uniqueness of the continuous extension. Finally,

$$|S_n(\Psi)| \leq c_n \sum_{\boldsymbol{\kappa}} |c_{\boldsymbol{\kappa}}(\Psi)| \leq 2\sqrt{n!} \left(\frac{34\pi\sqrt{6}}{4} \right)^n q_{6n}(\Psi),$$

which implies (32). □

7. Application to the QED3 curvature

The lattice input is a probability law of real curvature. For the all-flux chamber model, the infinite-volume construction in [Ch, Theorem A(i) and Section 2] gives such a law at each fixed spacing. Its finite model has antiperiodic spinors, the chamber fermion operator, and parameters

$$q > 0, \qquad 0 < \rho < 1, \qquad w_0, w_1, w_2, w_3 > 0, \qquad M \in \mathbb{R}.$$

The photon curvature is $u = dA + (2\pi/q)z$, with all closed integer fluxes z included. The precise operator and photon measure are those of the cited section. The parameters may depend on the spacing but are fixed during each volume limit; the physical charge scaling is $q_a = e\sqrt{a}$ for fixed $e > 0$.

Every selected real-curvature limit ν_a from that construction satisfies

$$\mathbb{E}_{\nu_a} \exp\left(\sum_{x,i<j} b_{ij}(x)u_{ij}(x)\right) \le \exp\left(\frac{1}{2}\sum_{x,i<j} |b_{ij}(x)|^2\right) \tag{35}$$

for every finitely supported real b. The constant is one, independently of the spacing and the selected volume limit. For completeness, the passage from finite volume uses only a fixed finite support: it embeds injectively into sufficiently large tori, so its counting norm is unchanged. The finite Gaussian estimate [Ch, Proposition 5.1] applies there. Weak convergence passes the bounded continuous cylinder function $\min\{R, \exp(\sum b_{ij}u_{ij})\}$ to the limit, and monotone convergence as $R \to \infty$ gives (35).

7.1. **Centered plaquettes and their continuum limits.** An oriented lattice two-cochain has real components $u_{\ell m}(x)$, $x \in \mathbb{Z}^3$, $\ell < m$, with the usual sign under reversal of orientation. Its coboundary is

$$(du)(x) = u_{23}(x+e_1) - u_{23}(x) - u_{13}(x+e_2) + u_{13}(x) + u_{12}(x+e_3) - u_{12}(x).$$

For $p = (x; \ell m)$ and $a > 0$, set

$$Q_a(p) = a(x + (e_\ell + e_m)/2 + [-1/2, 1/2)^3), \qquad f_{[a]}(p) = a^{-3/2}\int_{Q_a(p)} f_{\ell m}(y)\,\mathrm{d}y. \tag{36}$$

The ℓm component of the step field F_a equals $a^{-3/2}u_{\ell m}(x)$ on $Q_a(p)$. The geometric action is the exterior-square action on two-forms.

Corollary 7.1. Let $a_n > 0$ tend to zero. For each n, let ν_{a_n} be a Borel probability law on real arrays $(u_{ij}(x))_{x\in\mathbb{Z}^3,\,1\le i<j\le 3}$ satisfying (35). Let F_{a_n} be their centered-plaquette step fields as in (36). There are a subsequence and a Borel probability law μ of real tempered two-forms on $\mathbb{R}^3$ such that the laws of these step fields converge weakly to μ in H_{-4}. Here

$$\|F\|_{-4}^2 = \sum_{i<j}\sum_{k\in\mathbb{N}_0^3} (2|k|_1 + 4)^{-8} |F(h_k\,\mathrm{d}y_i \wedge \mathrm{d}y_j)|^2.$$

Each step-field law and every such limit μ satisfy

$$\mathbb{E}e^{F(f)} \le e^{\|f\|_2^2/2} \qquad (f \in \mathcal{S}(\mathbb{R}^3; \Lambda^2) \text{ real}).$$

Along the selected subsequence, all joint moments and cumulants of complex Schwartz pairings converge to those of μ. Their finite-family generating functions and logarithms have the locally uniform convergence asserted in theorem A. In particular, this conclusion applies to any choices of the infinite-volume real-curvature laws of the chamber model just described. ■□

Proof. For a finitely supported b, let $Y = \sum b_{ij}(x)u_{ij}(x)$ and $v^2 = \sum |b_{ij}(x)|^2$. Substituting tb in (35) gives $\mathbb{E}e^{tY} \le e^{t^2v^2/2}$ for every real t. As in [Cur, Proposition 5.1], this implies

$$\mathbb{E}|Y|^2 \le v^2. \tag{37}$$

The one-variable calculation is the first paragraph of the proof of proposition 2.1.

Apply proposition 3.1 to each law with $d = r = 3$, $C = 1$, the three components $12, 13, 23$, and their offsets $(e_i + e_j)/2$. Then apply proposition 4.1 to the sequence of step fields with $s = 3$ and $t = 4$, as in the proof of theorem B. This also permits different choices of law when a spacing occurs more than once. The resulting weak limit on H_{-4} has exactly the two-form interpretation and norm stated above.

It remains to retain the exponential estimate. For a real Schwartz two-form f, let $b_{a_n}^{(m)}$ be the finite cell-coefficient truncations used in proposition 3.1. Their pairings converge almost surely to $F_{a_n}(f)$. Fatou's lemma and (35) give

$$\mathbb{E}e^{F_{a_n}(f)} \le \liminf_{m\to\infty} \exp(\|b_{a_n}^{(m)}\|_{\ell^2}^2/2) = \exp(\|b_{a_n}\|_{\ell^2}^2/2) \le e^{\|f\|_2^2/2}.$$

This is the infinite-lattice test extension of [Lat, Section 6.1]. Apply theorem A to the weakly convergent subsequence already chosen. It gives the Gaussian bound for μ and all the stated correlation and cumulant limits. □

The following consequence adds the lattice identities and reflection hypotheses. The preceding corollary supplies weakly convergent subsequences; the assertion below applies to every selected weak limit.

Corollary 7.2. Let $a_j \downarrow 0$ and let ν_j be laws of real two-cochains on $\mathbb{Z}^3$. Suppose that:

(i) $du = 0$ almost surely and $\mathbb{E}_{\nu_j} e^{\langle h,u\rangle} \le e^{\|h\|^2_{\ell^2}/2}$ for every finitely supported real plaquette test h.
(ii) Each ν_j is invariant under integer translations, the oriented cubic action and $u \mapsto -u$.
(iii) Each ν_j is reflection positive for $x_\ell = 0$ and $x_\ell \pm x_m = 0$, $\ell < m$, on character sums supported on plaquettes all of whose corners lie strictly on the chosen positive side.
(iv) The laws of F_{a_j} are realized on H_{-4} and converge weakly there to P.

Then P is a translation-invariant law of closed real tempered two-forms, invariant under the cubic group and field-sign change. It satisfies $\mathbb{E}_P e^{F(f)} \le e^{\|f\|_2^2/2}$ and is invariant and reflection positive at every plane $x_\ell = c$, $x_\ell \pm x_m = c$, $c \in \mathbb{R}$. All fixed finite-family moments and cumulants converge along the given sequence. ■□

7.2. **Pairings and equivariance.** Proposition 3.1 gives the measurable realization and $F_a(f) = \langle f_{[a]}, u\rangle$. The cell estimate (22) and the Fatou argument in corollary 7.1 give

$$\mathbb{E} e^{F_a(f)} \le e^{\|f\|_2^2/2}. \tag{38}$$

If γ is an affine cubic symmetry of the unit lattice and γ_a its physical rescaling, define $(\gamma \cdot h)(p) = h(\gamma^{-1}p)$, where reversing the orientation of a plaquette reverses the coefficient's sign. A change of variables gives

$$(\gamma_a \cdot f)_{[a]} = \gamma \cdot f_{[a]}. \tag{39}$$

Signed coordinate permutations carry cell centers to the corresponding plaquette centers and carry cubes to cubes, up to null boundaries. The orientation sign is the exterior-square matrix in both actions. Integer translations give physical translations in $a\mathbb{Z}^3$. These are the exact symmetries used in section 5.

7.3. **The discrete identity and the reflection support.** Let $\xi \in C_c^\infty(\mathbb{R}^3;\mathbb{R})$ denote the three-form $\xi\, \mathrm{d}y_1 \wedge \mathrm{d}y_2 \wedge \mathrm{d}y_3$. Define

$$\partial^a_\ell \xi(y) = a^{-1}(\xi(y + ae_\ell/2) - \xi(y - ae_\ell/2)), \qquad (v^a_{23}, v^a_{13}, v^a_{12}) = (-\partial^a_1\xi, \partial^a_2\xi, -\partial^a_3\xi).$$

Then

$$F_a(v^a) = 0, \qquad \|v^a - d^*\xi\|_2 \le \frac{a^2}{24}\sum_{\ell=1}^3 \|\partial^3_\ell \xi\|_2. \tag{40}$$

For the exact identity, assign to each unit cube the number

$$\xi_{[a]}(x) = a^{-5/2}\int_{a(x+[0,1)^3)} \xi(y)\,\mathrm{d}y.$$

The $23, 13, 12$ coefficients of $v^a_{[a]}$ are

$$-(\xi_{[a]}(x) - \xi_{[a]}(x - e_1)), \quad \xi_{[a]}(x) - \xi_{[a]}(x - e_2), \quad -(\xi_{[a]}(x) - \xi_{[a]}(x - e_3)).$$

Their finite pairing with u regroups to $\sum_x \xi_{[a]}(x)(du)(x) = 0$. The Fourier multiplier of $\partial^a_\ell - \partial_\ell$ is $i(2a^{-1}\sin(ap_\ell/2) - p_\ell)$, whose modulus is at most $a^2|p_\ell|^3/24$. Plancherel gives the second assertion.

For a mirror with defining function $\ell(y) = y_j$ or $y_j \pm y_k$, take compact tests supported in $\{\ell \ge \delta\}$, $\delta > 0$. A nonzero cell coefficient comes from a cell meeting this set. For every corner q of its physical

plaquette and every y in that cell, $|\ell(q)-\ell(y)| \le 2a$: each coordinate displacement is at most a, and the normal has at most two nonzero coordinates of modulus one. Thus $2a < \delta$ puts all contributing plaquette corners strictly on the positive side. There are only finitely many such plaquettes. The lattice reflection inequality and (39) give (12).

Proof of corollary 7.2. Equation (38) gives the common Gaussian input. Equation (40) verifies the identity hypothesis with $D^* = d^*$. Equivariance gives cubic symmetry and mesh translations converging to every continuum translation. The support check gives the eventual character inequalities at the nine origin mirrors. Theorem C applies. Field-sign invariance passes directly through its characteristic-function identity. □

The real-curvature laws supplied by [Ch, Theorem A] satisfy the additional hypotheses of corollary 7.2. Together with corollary 7.1, this gives subsequential continuum laws with all the stated structural properties. The reflection passage is the origin-height argument of [Cur, Proposition 6.1]; translations supply every height. The conditional corollary imposes no arithmetic restriction on the spacing sequence.

7.4. **Recovery of a prescribed volume limit.** An infinite-lattice law gives a step-field law by proposition 3.1. To treat a prescribed volume limit of the finite-torus fields, we also need the converse identification. We use the bump-function argument of [Cur, Section 6.4].

Fix a spacing $a > 0$. Let $\mathcal{P}_2$ be the positively oriented plaquettes of $\mathbb{Z}^3$, and put

$$\Omega = \{u \in \mathbb{R}^{\mathcal{P}_2} : du = 0\},$$

with its coordinate product topology. For each integer $n \ge 4$, let $\lambda_{a,n}$ be a probability law of real closed two-cochains on $(\mathbb{Z}/n\mathbb{Z})^3$ satisfying

$$\mathbb{E}_{\lambda_{a,n}} e^{\langle \beta, u\rangle} \le e^{\|\beta\|^2_{\ell^2((\mathbb{Z}/n\mathbb{Z})^3)}/2} \qquad (\beta \text{ a real torus two-cochain}). \tag{41}$$

Both the pairing and the norm sum over positively oriented torus plaquettes. Denote the law of the periodic extension by $\nu_{a,n}$. Use the centered interpolation and coefficients $f_{[a]}$ of (36), and write $P_{a,n}$ for its law on H_{-4}. For general Schwartz tests these finite-torus laws obey the periodized bound (47) below, with physical period $R = an$. The following identification gives the sharp L^2 bound after the volume limit.

Proposition 7.3. Suppose $n_j \to \infty$ and $P_{a,n_j} \Longrightarrow \mu_a$ weakly on H_{-4}. Then there is a probability law ν_a on Ω such that

$$\nu_{a,n_j} \Longrightarrow \nu_a, \qquad \mu_a = (\sharp_a)_* \nu_a, \tag{42}$$

where $\sharp_a$ is the centered step map, defined as a Borel map into H_{-4} on a set of full ν_a measure and set equal to zero elsewhere. The convergence of cochain laws is in the product topology and holds along the given sequence. For every finitely supported real cochain β,

$$\mathbb{E}_{\nu_a} e^{\langle \beta, u\rangle} \le e^{\|\beta\|^2_{\ell^2(\mathcal{P}_2)}/2}. \tag{43}$$

In particular, μ_a satisfies the Gaussian bound $\mathbb{E}_{\mu_a} e^{F(f)} \le e^{\|f\|_2^2/2}$ for real Schwartz two-forms f. ■

Proof. Discard finitely many terms so that $an_j \ge 1$. The finite-volume step maps take values in H_{-3}. Indeed, the periodization estimate below and the Hermite bound $p_1(h_k e_\alpha) \le 31\lambda_k$ from [26, Section 3] give $|F_a(h_k e_\alpha)| \le 31\lambda_k \|u\|_{\ell^2}$. Summing with weight λ_k^{-6} proves continuity of each finite-dimensional step map into H_{-3}, since $\sum_k \lambda_k^{-4} < \infty$.

We use the realization already proved in proposition 3.1. Let Ω_3 be its weighted square-summability domain with exponent $p = 3$, intersected with Ω. For every $u \in \Omega_3$, the step distribution has the absolutely convergent pairings (19). Let $\Omega^{(a)} \subset \Omega_3$ be the set on which it has finite H_{-3} norm, and set $\sharp_a = 0$ off this set. The measurability argument in proposition 4.1 makes $\Omega^{(a)}$ measurable and

$\sharp_a$ a Borel map into H_{-4}. Every periodic cochain lies in Ω_3 and, by the preceding finite-volume estimate, in $\Omega^{(a)}$.

For each plaquette σ, choose a real smooth bump b^σ with only its corresponding two-form component nonzero, supported inside $Q_a(\sigma)$ and of integral $a^{3/2}$. Then $(b^\sigma)_{[a]} = \mathbf{1}_\sigma$. The map

$$\mathcal{J}_a : H_{-4} \longrightarrow \mathbb{R}^{\mathcal{P}_2}, \qquad \mathcal{J}_a(F)(\sigma) = F(b^\sigma), \tag{44}$$

is continuous for the product topology, and $\mathcal{J}_a \sharp_a u = u$ on $\Omega^{(a)}$. At finite volume the identities are

$$P_{a,n_j} = (\sharp_a)_* \nu_{a,n_j}, \qquad \nu_{a,n_j} = (\mathcal{J}_a)_* P_{a,n_j}.$$

The mapping theorem therefore gives $\nu_{a,n_j} \Longrightarrow \nu_a := (\mathcal{J}_a)_* \mu_a$. The closed conditions $du = 0$ pass to this product limit.

Fix a finitely supported real β on the covering lattice. For all sufficiently large n_j, its support embeds injectively in the torus. Its folded norm then equals its covering ℓ^2 norm. Apply (41), pass the bounded continuous cylinder $\min\{R, e^{\langle \beta, u\rangle}\}$ to the limit, and let $R \to \infty$. This proves (43).

The Gaussian bound implies the finite-coefficient second-moment bound, so proposition 3.1 gives $\nu_a(\Omega_3) = 1$. The step distribution on this domain satisfies the common L^2 test estimate. Applying proposition 4.1 with $s = 3$ therefore gives $\nu_a(\Omega^{(a)}) = 1$. The finite-coefficient Fatou argument in corollary 7.1 now gives

$$\mathbb{E}_{\nu_a} e^{(\sharp_a u)(f)} \le e^{\|f_{[a]}\|_{\ell^2}^2/2} \le e^{\|f\|_2^2/2}.$$

It remains to prove the reverse equality in (42). Choose a countable Schwartz-dense family $\mathcal{D}_0$ of real compact smooth two-forms. For $f \in \mathcal{D}_0$, the test

$$f' = f - \sum_\sigma f_{[a]}(\sigma) b^\sigma$$

is a finite sum and satisfies $f'_{[a]} = 0$. Every finite-volume step field therefore annihilates f'. The hyperplane $\{F : F(f') = 0\}$ is closed in H_{-4} and has full μ_a measure. Intersect these events for $f \in \mathcal{D}_0$ with the full-measure event $\mathcal{J}_a(F) \in \Omega^{(a)}$. On that single event,

$$F(f) = \sum_\sigma f_{[a]}(\sigma) F(b^\sigma) = (\sharp_a \mathcal{J}_a F)(f) \qquad (f \in \mathcal{D}_0).$$

Both sides are tempered distributions, so density gives $F = \sharp_a \mathcal{J}_a F$ there. Taking laws proves the pushforward identity and the Gaussian assertion for μ_a. □

7.5. **Finite-torus estimates and joint cutoff sequences.** The exact L^2 Gaussian bound in theorem A applies to the recovered infinite-volume laws. Before the volume limit, the Schwartz test is periodized. We recall the estimate from [Cur, Sections 6.1–6.2] that controls this step.

Put $R = an$ and $f^R(y) = \sum_{k \in \mathbb{Z}^3} f(y + Rk)$. The finite-torus coefficient is

$$b_{a,n}([\sigma]) = \sum_{k \in \mathbb{Z}^3} f_{[a]}(\sigma + nk) = a^{-3/2} \int_{Q_a(\sigma)} f^R_{\ell m}(y)\, \mathrm{d}y \quad (\sigma = (x; \ell m)). \tag{45}$$

The sum converges absolutely. It gives $F_a(f) = \langle b_{a,n}, u\rangle$, and cell Cauchy–Schwarz gives $\|b_{a,n}\|_{\ell^2} \le \|f^R\|_{L^2(\mathbb{R}^3/R\mathbb{Z}^3)}$. With $Q_R = [-R/2, R/2)^3$, define

$$p_R(f) = \|f\|_2 + 26 R^{-2} \||y|^2 f\|_{L^2(\mathbb{R}^3 \setminus Q_R)}. \tag{46}$$

Then

$$\mathbb{E}_{P_{a,n}} e^{F(f)} \le e^{\|b_{a,n}\|_{\ell^2}^2/2} \le e^{p_R(f)^2/2} \qquad (f \text{ real}). \tag{47}$$

To see the last estimate, put $c_k = \|f\|_{L^2(Q_R + Rk)}$. Minkowski gives $\|f^R\|_{L^2(\mathbb{R}^3/R\mathbb{Z}^3)} \le \sum_k c_k$. For $k \neq 0$, $|y| \ge R|k|_\infty/2$ on $Q_R + Rk$, and weighted Cauchy–Schwarz gives

$$\sum_{k \neq 0} c_k \le 4R^{-2} \left(\sum_{k \neq 0} |k|_\infty^{-4} \right)^{1/2} \||y|^2 f\|_{L^2(\mathbb{R}^3 \setminus Q_R)}.$$

The shell count $24j^2 + 2$ gives $\sum_{k\neq 0} |k|_\infty^{-4} = 4\pi^2 + \pi^4/45$, whose square root times four is less than 26. For $R \geq 1$, one has $p_R(f) \leq p_1(f)$, and $p_R(f) \to \|f\|_2$ as $R \to \infty$. The seminorm p_1 is continuous on Schwartz space.

Corollary 7.4. At each spacing $a_i \downarrow 0$, take the chamber torus laws with parameters fixed during that volume limit. Let μ_i be any prescribed weak H_{-4} volume limit of their centered step fields. Suppose $\mu_i \Longrightarrow \mu$ on H_{-4}. Then μ has the conclusions of corollary 7.2, including convergence of all finite-family moments and cumulants from μ_i to μ. Its tensor Schwinger functions satisfy proposition 6.1. There are torus sides $n_i \to \infty$ such that $a_i n_i \to \infty$,

$$P_{a_i,n_i} \Longrightarrow \mu,$$

so μ is also realized by a joint sequence of finite cutoff laws. ■□

Proof. Apply proposition 7.3 at each a_i. It identifies the prescribed law μ_i with the step image of a weak cochain limit. The chamber theorem [Ch, Theorem A] applies to every such limit with the parameters fixed: it gives closedness, Gaussian domination, the lattice symmetries, and reflection positivity at the nine mirror families. Thus these recovered laws meet all the hypotheses of corollary 7.2; that corollary and proposition 6.1 give the assertions about μ and the sequence μ_i.

Fix a metric for weak convergence of probability laws on H_{-4}. Along the defining volume sequence for μ_i, choose n_i so large that $a_i n_i \geq i$ and the distance between P_{a_i,n_i} and μ_i is less than $1/i$. Then $P_{a_i,n_i} \Longrightarrow \mu$ and $R_i := a_i n_i \to \infty$. This is the diagonal selection in [Cur, proof of Proposition 6.1]. □

References

[26] "The continuum field-strength construction for QED in three dimensions". Modular revision of 20 October 2026. 2026. URL: https://github.com/ethanwsussman/Yang--Mills-LLM-experiment/blob/main/QED3/QEM1.pdf (cit. on p. 13).

[Ch] "Reflection positivity for chamber QED on the infinite lattice". Companion manuscript, Theorem A and Proposition 5.1. 2026. URL: https://github.com/ethanwsussman/Yang--Mills-LLM-experiment/blob/main/QED3/Infinite-lattice%20reflection%20positivity%20%28LP1%29/LPM1.pdf (cit. on pp. 4, 10, 11, 13, 15).

[Cur] "The field-strength Osterwalder–Schrader axioms for QED in three dimensions". Comprehensive edition, 263 pp., 20 October 2026, before the modular integration revision. 2026. URL: https://github.com/ethanwsussman/Yang--Mills-LLM-experiment/blob/main/QED3/qed3.pdf (cit. on pp. 2–4, 7, 9, 11, 13–15).

[Lat] "Subsequential infinite-volume limits for lattice gauge fields". Earlier revision, including the real-curvature and continuum sections. 2026 (cit. on pp. 3, 7, 12).

[RT07] R. Radha and S. Thangavelu. *Holomorphic Sobolev spaces, Hermite and special Hermite semigroups and a Paley–Wiener theorem for the windowed Fourier transform*. 2007. arXiv: 0710.3481 (cit. on pp. 3, 7).

[Sor24] E. Sorensen. *TA session: tightness of probability measures*. SLMath Stochastic Quantization Summer School. 2024. URL: https://www.mit.edu/~skycao/SLMath/Session_1_notes_(tightness).pdf (cit. on pp. 3, 8).